Renewable and Sustainable Energy Reviews 189 (2024) 114029

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Local acceptance of solar farms: The impact of energy narratives

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ARTICLE INFO

Keywords:
Solar farm
Local acceptance
Social licence
Network model
Energy transition
Narratives

ABSTRACT

This research explores how a new concept, support for energy transition narratives, and other commonly applied social licence factors influence local acceptance of large-scale solar farms. Producing renewable electricity requires deploying more large-scale renewable technology infrastructure and support of host communities will be paramount. There is a dearth of studies that investigate a comprehensive set of social licence factors that influence local support of solar farms. Moreover, there is a lack of research considering how broader environmental worldview and beliefs about the energy transition influence local level support for renewable energy infrastructure. Survey data from a large, representative sample of the Australian population is analysed using a network modelling approach to identify conditional associations between variables of interest. Results reveal that general support for renewables in the energy transition had relatively strong direct and indirect associations with local acceptance in addition to social licence factors. Conversely, beliefs about the energy transition beyond the role for renewables (i.e., support for the role of natural gas, support for exporting fossil fuels and willingness to make trade-offs for a faster energy transition) had relatively weak effects on local acceptance and other social licence factors. The findings suggest that individual alignment with broader energy transition narratives explains local solar farm support beyond social licence factors. Future social licence research should consider the role of broader public narratives when investigating the factors that facilitate or impede local acceptance of renewable energy projects.

1. Introduction

Many countries around the world are transitioning to lower emission energy technologies. One key component of this energy transition is the use solar of photovoltaics to create electricity [1]. Solar energy (e.g., using solar photovoltaics) is one of the fastest growing energy production methods and will have an important role to play in reducing the world's reliance on fossil fuels [2]. However, producing energy at the scale required to replace fossil fuel-based production methods will require large scale solar farms, in addition to roof-top solar. It is, therefore, important to understand the factors that may facilitate or impede the successful development and deployment of large-scale solar farms.

One important stakeholder to consider when building large-scale solar farms is the community that will live near them. Research has shown that community acceptance of public siting energy technologies, or a lack thereof, can greatly influence the successful deployment of renewable energy projects [3]. Therefore, it is important to understand, ahead of time, the factors that may influence the successful implementation of large-scale solar farms. Unlike other energy production technologies like wind turbines, few studies have comprehensively explored the acceptance of large-scale solar farms [4]. As such, this research explores the community acceptance of large-scale solar farms associated with seventeen explanatory factors (common social licence factors, worldviews, and agreement with the energy transition narrative) using a large, representative sample of the Australian population. Moreover, this work uses a novel exploratory data analysis approach to identify potential causal mechanisms that influence local community support for solar farms. Understanding these factors can inform planning and decision-making including approaches to communication and engagement strategies and other interventions that support responsible development and deployment of large-scale renewable energy infrastructure into local communities.

1.1. Wider public attitudes versus local community acceptance

Research to date suggests that people are generally supportive of using renewables and solar power in particular [5–7]. Visschers and

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https://doi.org/10.1016/j.rser.2023.114029
Received 10 January 2023; Received in revised form 30 September 2023; Accepted 26 October 2023
Available online 15 November 2023
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Siegrist, for example, found that people tend to be more accepting of solar power than other energy technologies such as nuclear, wind and natural gas [6]. Similar results have also been found in Australia, with people tending to support the use of wind and solar over other energy technologies like hydro-power, nuclear, coal and gas [5]. Another study comparing levels of renewable energy acceptance across countries in Europe also found that solar power had the highest level of support of all renewable energy technologies [8]. Overall, these findings suggest that people, for the most part, like the idea of using more solar power in the future; though it is less clear on people's perceptions of large scale solar farms.

Furthermore, high levels of broader public acceptance does not necessarily translate into high levels of acceptance at the smaller community level [9–13]. Some community members experience, or expect to experience, more of the costs (e.g., local environmental impacts) and less of the benefits (e.g., wealth creation) associated with an energy production technology than the broader population [12]. It is not uncommon for publicly supported renewable energy projects to be met with local opposition [14]. Wind farms, for example, are strongly supported at the population level but often face local opposition for a variety of reasons [15,16]. Few studies have, though, comprehensively investigated how communities think about, and are likely to respond to, large-scale solar farms [2,4,17], especially the role of the wider energy transition narrative on local community acceptance.

1.2. Local community acceptance of renewable energy developments

Recent empirical research has investigated levels of local support for utility-scale solar energy installations and the factors that influence it. Carlisle and colleagues in a series of studies found that local support for utility-scale solar in the US is high, but varies depending on a number of factors [2,17,18]. In two studies, they found that local support for utility-scale solar in California was significantly influenced by perceived positive impacts like jobs and increased property value [2,18]. They also found that, contrary to past research [11], place-based beliefs (e.g., level of agreement that where they lived was beautiful, had a great community of people etc.) and perceived negative impacts had no significant influence on support for utility-scale solar when controlling for other factors like perceived positive impacts [2,18]. In another study, using a sample from the whole US population, it was found that political ideology, believing that a solar farm would spoil scenery, and trust in the developers were significant predictors of acceptance of a solar farm in one's local county [17]. Other factors like the perceived seriousness of climate change and believing that a solar development would decrease property value were not significant predictors of local acceptance [17]. These findings suggest the perceived personal and community benefits of a local solar project are particularly strong drivers of local acceptance. However, the statistical models in all these studies only explained relatively small amounts of variability (between 18 and 40 % in sample) in local support, which suggests other important factors need to be included to more comprehensively explain solar farm acceptance.

More recent research investigating local acceptance suggests that while communities are not against large-scale solar, they are more supportive of smaller scale solar applications [4,19]. Two recent studies, for instance, found that support for large 'utility-scale' solar was significantly lower than support for rooftop or community-scale solar [4,19]. The findings suggest that as the size of the projects increase, the level of support tends to decrease. Cousse also found that, contrary to past findings, people express similar levels of support for solar and wind when controlling for the size of the project suggesting that the type of technology is less important than the overall footprint [4]. One explanation for this finding is that larger-scale solar projects seem to be viewed as an 'industrial activity' and thus people tend to think more about the local impacts [19].

Theories of social licence and reviews of empirical research highlight a few key factors that help to explain local acceptance of energy-related projects [10,20–22]. These beliefs and attitudes can be broadly categorised into three main explanatory factors: (1) beliefs about a project's consequences (2) social trust and relationships with industry operators, and (3) implicit attitudes and/or feelings about the energy technology. This section will briefly summarize these three main belief/attitude categories and highlight how they are usually operationalised in empirical research.

One important set of beliefs that is a reliable predictor of local acceptance is how people think about the consequences (i.e., effects) of an energy project. Perceived consequences are often categorised as the perceived negative impacts (e.g., perceived risks and/or costs) and the perceived benefits (both local and societal). Studies that measure perceived impacts (or perceived risk/cost) find that the more people agree that a project will cause negative outcomes, the less they support it [23–28]. Conversely, believing that an energy project will bring about benefits is positively related to acceptance [29–33]. There is some evidence to suggest that perceived individual and community effects of a solar farm (both benefits and impacts) are also important for explaining local solar farm acceptance [2,18]. However, the past research investigating large scale solar farms uses relatively few survey items to assess perceived impacts and benefits. As such, it is relatively unclear from previous research what types of perceived effects are important for explaining solar farm acceptance and why.

Of particular importance is the extent to which people trust the company involved with an energy project or the development of a technology [20,34]. Several studies have shown that social trust is strongly associated with acceptance in many different contexts [26,35–37], including the acceptance of solar farms [17]. However, the findings are mixed regarding the extent to which social trust explains acceptance in less contentious energy industries like renewable energy projects [9,23]. Associated with social trust, other psychosocial factors also seem to explain local acceptance of energy projects. For example, procedural fairness, believing that the community has been fairly involved in a company's decision making, and relationship quality, believing that there is a good community relationship with a company, have both been shown to be strongly association with trust and, in turn, local acceptance [26,35]. In addition to these factors, other research finds that distributive fairness (i.e., believing that local benefits will be shared fairly) and governance (i.e., believing that governance would be effective and trustworthy) also help to explain variability of a measure of local acceptance [26]. While these factors do seem to explain local acceptance in contentious energy industries (e.g., mining and coal seam gas extraction) it is unclear from past research how much these additional contextual factors help to explain local solar farm acceptance.

Beyond more explicit beliefs about a project's impacts and the context, affect and/or feelings tend to be another reliable predictor of energy technology acceptance. Affect is a broad term that can encompass general moods, specific emotions and feelings.[1] Psychological research has consistently shown that people's affect and/or feelings are strongly associated with explicit attitudes, preferences and behaviour [38,39]. The affect heuristic provides one explanation for this finding and posits that people tag attitude objects (people, places, technology) with either positive or negatively valanced imagery [40,41]. Put simply, people tend to form quick evaluations of things based on feelings of relative 'goodness' or 'badness' they associate with it.

There is some evidence to suggest the feelings people associate with solar farms help to explain their reported level of acceptance. For example, Cousse found that one of the strongest predictors of local concern about a large-scale solar farm was the affective responses people report (i.e., 'liking' or 'disliking') when shown pictures of a solar farm [4]. The same study found that whilst people tended to think of

[1] Affect is sometimes uses synonymous with the concept of an *attitude*. That is, an explicit (or implicit) liking or disliking of an attitude object. The word *feeling* instead of *affect* is used in this research to differentiate it from *attitude*.

positively valanced words when asked to think about solar panels in general, when respondents were presented with pictures of large-scale solar farms they tended to think of more negatively valanced words. The results suggest people tend to associate images of larger scale projects with more negatively imbued thoughts, which in turn, lowers community acceptance [4]. These findings are supported by other research showing that people tend to think about solar power, and other renewables, in an abstract way, which tends to lead to more favourable evaluations [9]. However, when people are asked to think more about the potential negative impacts of a solar farm project (e.g., toxic waste generated during production), then acceptance is lower [9]. The findings suggests that the feelings people associate with solar farms should help to explain acceptance.

1.3. Wider public attitudes toward the environment and energy

While most studies investigating social acceptance of energy technologies tend to focus on contextual factors, other research suggests that broader beliefs people hold about related topics (e.g., climate change, the environment, politics) help to shape how people think about energy projects [42]. Several studies have, for example, shown that pro-environmental beliefs (and/or environmental concern) and specific values (e.g., altruism) help to shape attitudes towards a range of energy technologies [16,43–46]. Moreover, these values seem to influence acceptance through their influence on more context specific beliefs like the perceived risk and benefit of a technology [16,29,42,47].

There is evidence to suggest that broader beliefs and values also influence local solar farm acceptance. Carlisle et al. highlights that broader beliefs about the environment and one's political affiliation are related to local support for utility-scale solar [17]. Cultural theory posits that environmental values, beliefs, and attitudes interplay with political perspectives and about how best to address environmental issues such as climate change [48]. This interplay is reflected in broad worldviews about relationships between humans and nature, the robustness of the environment to change, the need for human intervention, and the role of government and technology [49].

Another important aspect that has received little consideration in past research is the extent to which people support more specific narratives about the renewable energy transition [50,51]. A narrative typically addresses why a technology is needed, why this technology over other options, and how this technology solution fits into a broader solution, in this case as part of a contested energy mix [50]. The narrative reflects the underlying rationale as to how effectively the technology addresses a broader problem or issue. Narratives have been identified as important in shaping public debate and critical in speeding up or slowing down a transition to novel technologies [52].

The narrative around the energy transition developing in Australia has different dimensions based on the role that individual energy sources are seen to play in the economy and how they play into the trilemma of energy needs [53,54]. Australia, for example, is a country that has created much of its wealth due to mineral and energy extraction and export. As such, energy transition narratives include implications for these industry sectors. The energy narrative also stipulates a need for energy to be affordable, sustainable and reliable for the everyday household and for the functioning of the economy. This implies a potential trade-off: the willingness of households to trade-off energy affordability and reliability for a speedier energy transition.

1.4. A country experiencing an energy transition

The public discussion of the energy mix of the future also needs to be set in context of the role that energy plays in the revenue of the country and national economic benefits. Australia is an energy exporter with coal mining accounting for 1.7 % and activities from oil and gas extraction accounting for around 2 % of the gross domestic product [55] [2]. Additionally, royalties contribute around $10 billion or 4.4 % of the states' general finance statistics revenue [55].[3] Almost 19 % of Australia's product exports in 2019 were fuels [56] and the country is one of the top four global exporters of LNG and a top two coal exporter [57]. Like many countries dependent on fossil fuel related revenues, a transition to the use of renewables and phasing out of fossil fuels extraction can be perceived as a risk for this source of revenue and economic activity in the country. It is, therefore, likely that public beliefs about the broader energy system in Australia and, particularly its historical reliance on the fossil fuel industry for economic growth, will influence how people think about renewables and, in turn, solar farms.

Different energy resources will have different timelines for being phased in or phased out. In global demand scenarios based on stated policies and announced pledges, natural gas still appears to have a long-term role to play in the energy mix of countries and plausibly in Australia's energy exports [58]. Perceptions about natural gas and its future role in Australia [53], as well as the country's capacity to become an exporter of renewable energy should also influence views about the role for renewables and local acceptance of solar farms. Determining the extent to which such view influence local acceptance also has implications for other countries outside of Australia experiencing a similar energy transition who also rely on natural gas as part of their energy system.

1.5. Aims

This study has two main aims. The first is to provide a descriptive representation of the social licence factors that are associated with local large-scale solar farm acceptance. Past research highlights the importance of explanatory factors such as perceived impacts, benefits and social trust for explaining local acceptance of energy technologies. However, the relative importance of such factors, using a comprehensive set of measures, for explaining the local acceptance of solar farms has yet to be tested empirically. The second aim of this study is to explore the extent to which broader beliefs about the energy transition narrative may further influence local solar farm acceptance (directly and via other explanatory factors). As this study uses an exploratory data analysis approach, no specific predictions are made about the relative importance of specific variables, causal mechanisms, or the validity of a specific theoretical model. Instead, the goal of this study is to identify robust effects in a large, representative sample to aid future research and theory development. The findings are also relevant to practitioners seeking to understand the key factors underpinning social acceptance of large-scale solar farms and how these are influenced by energy transition narratives. The research is of particular relevance to those parts of the world where fossil fuel related industries still hold a key role in the economy.

2. Methodology

This research used an online survey instrument to elicit a range of

[2] On an industry level gross value-added basis. Figures approximate contributions for the most recent years of data available, namely FY 2016–17 and 2017–18. According to the same measure the mining sector overall accounts for around 9.5 % of the national gross domestic product.

[3] Average over 2011–2020 for all royalties paid at state level. The most recent year of data is the FY 2019–20 where state level royalties reached more than $15 billion and accounted for about 5.6 % of the states' government revenue.

beliefs and attitudes towards the development of a local solar farm. In survey research, the goal is to obtain a representative sample of target population to make reliable inferences about that population. This research used an undirected network modelling approach to determine the conditional associations between a number of theoretically relevant social licence constructs.

2.1. Participants

Respondents for this study were recruited using an online research panel to obtain a representative sample of the Australian population. To recruit participants, a market research company sent information about the survey to Australians enrolled on their research panel. Participation in the study was voluntary for members of the research panel and participaptns were awarded points (redeemable for monetary vouchers) from the market research company for participation. In total, 3103 respondents completed the survey, 45.9 % male 53.8 % female and 0.2 % who preferred not to disclose the gender or identified as non-binary. This sample size was chosen to achieve sufficient statistical power for the planned analyses, which extended beyond the analysis presented in this research. The sample was representative of the Australia population with respect to gender, age and location of residence (see Walton et al. [59] for more information regarding the demographic characteristics of the sample). Half the sample was currently employed (50.1 %) and over a third (35.6 %) of the sample owned solar panels at home. Most of the respondents completed high school (84 %) and over a third had a university degree (i.e., a bachelors and/or postgraduate degree; 37.6 %).

2.2. Materials and procedure

Before being presented with questions assessing attitudes and beliefs regarding solar farms, all participants were presented with a hypothetical scenario about a planned solar farm project. Information regarding the planned solar project was presented in a mock letter from a hypothetical company building the solar farm (see Walton et al. [59] for the letters presented to the participants). In all scenarios, participants were told about the plan the build a solar farm near their town of residence, 10 kms away from the nearest residential house. Some aspects of the solar farm were experimentally manipulated (a $3 \times 2 \times 2$ design) to test whether specific solar farm characteristics influenced acceptance or other psychological factors of interest. These experimental conditions had little or no effect on social acceptance or any of the other psychological constructs assessed in the survey [59]. As such, the experiment conditions were not included as confounding variables in the subsequent analyses.

Solar farm acceptance, underlying social licence factors, and other psychological factors of interest were assessed using an online survey instrument. The survey instrument was created as part of a larger research project so not all the items were used in this study (see Walton et al. [59] for a full copy of the survey instrument). Informed by the theory and empirical research, multiple items were created to assess a solar farm acceptance and a range of relevant social licence factors (e.g., perceived benefits, impacts and social trust). All these survey items were assessed using seven-point Likert scales, and scores on the items were summed and averaged to assess the factors of interest (see Table 1). Seven-point scales were chosen to increase the sensitivity of the measure (i.e., capture more nuanced responses) and to increase the variability of responses for the analysis [60]. Most of these scales used to assess social licence factors were informed by past research. Some items were reverse scored before summing so that higher scores reflected greater levels of a construct (e.g., higher scores on social acceptance reflects greater support for the proposed solar farm project). As shown in Table 1 all scales for solar farm acceptance and social licence factors all showed high internal reliability ($\alpha > 0.80$ for all scales).

Single item measures were included for four worldviews, as reflected in Steg and Sievers 'myths of nature' - ephemeral, tolerant, benign, and

Table 1
The social licence and worldview factors used in the study.

Factors	Items	Cronbach α	Example items
Social acceptance	4	.96	Agreement with tolerating, accepting, approving, or embracing the development
Procedural fairness	4	.90	Agreement with having opportunities to participate in decisions made by the developer; and them listening and respecting community opinions
Trust in industry	2	.93	Being able to trust the developer to act responsibly; and that they would do the right thing by the community
Trust in governance	6	.88	Agreeing that government would be able to hold solar farm companies accountable; and the government would listen to and respect community opinions
Relationship quality	2	.88	Agreeing that the developer would be open, honest, and transparent; and engage in genuine two-way dialogue
Distributive fairness	3	.83	Agreeing it would be fair to have this development in their local area if the local council were compensated accordingly; and if the regional benefits outweighed any impacts
Perceived local benefits	8	.94	Local employment; local business opportunities; and corporate support for local community activities
Perceived societal benefits	6	.94	Make the region more attractive to new business and industry; boosting the wider state economy; providing energy security for the state
Perceived impacts	10	.94	Dust and noise pollution during construction; reducing the visual attractiveness of the local landscape; impacts of transmission lines
Role for renewables	11	.93	Solar farms have an important role to play in transitioning my state to renewable energy; in reducing reliance on coal for domestic energy; increasing our energy supplies from renewables will make electricity more affordable
Role for gas	2	.87	There is a role for natural gas in transitioning to renewable energy supplies if it displaces coal for electricity generation in Australia; if it does not delay renewable energy development
Role for fossil fuel exports	2	.87	As long as Australia increasingly obtains its domestic energy supply from renewable energies, there is a role for exporting coal; there is a role for exporting gas
Willingness to make trade offs	2	.78	To enable Australia to move faster to renewable energies, willingness to pay more for electricity; to risk more electricity blackouts
Fragile	1	NA	Agreeing that the environment is fragile and will only be protected if there are large changes in human behaviour and society
Manageable	1	NA	Agreeing that the environment can be managed by the government and experts if there are clear rules about what is allowed
Adaptable	1	NA	Agreeing that the environment can adapt to changes and technology will solve environmental problems eventually
Unpredictable	1	NA	Agreeing that the environment is unpredictable and we can't control what happens
Feelings (positive)	4	.85	Feeling pleased, optimistic, angry (reverse coded), or worried (reverse coded) about this solar farm development

capricious [61]. There worldview measures were adapted for the Australian context and were called: fragile, adaptive, manageable, and unpredictable respectively. Demographic factors were not included in the analysis as they were, for the most part, unrelated to the psychological constructs assessed (see Ref. [59]).

This study also assessed factors that have not been included in past survey research. As discussed in the introduction, this study aimed to explore the extent to which broader beliefs about the energy transition influenced solar farm acceptance. As beliefs about the energy transition have not been comprehensively assessed in past research, 17 new items were created to assess these beliefs. To reduce the number of variables for subsequent analysis, a principal component analysis was used. The results of the principal component analysis suggested a four-factor solution. These factors were labelled 'role for renewables' in the energy transition, 'role for gas' in the transition, 'role for fossil fuel exports' during the transition, and 'willingness to pay' or trade-off for energy transition (see Ref. [59] for the item factor loadings and the phrasing of each items). All variables were treated, and therefore modelled, as continuous variables in the subsequent analysis.

2.3. Data analysis method

The data were analysed using undirected network models to explore the conditional associations between the variables of interest. While still relatively novel when applied to human data, the network approach has been used in other contexts to explain attitudes [62], psychopathology [63], personality [64] and intelligence [65]. Applied research has also used this approach to explain attitudes towards plastic [66], vaccine decisions [67], exercise behaviour [68], environmental related beliefs [69] and responses to the COVID-19 pandemic [70]. However, to the authors' knowledge, this is the first time that a network modelling approach has been used to explore how psychological factors influence local acceptance of an energy technology. This innovative approach can shed light on the complex factors that drive acceptance of energy technologies, which is crucial for effectively implementing sustainable energy solutions in communities around the world.

Estimating undirected network models was selected as an analysis approach for two main reasons. First, network modelling is well suited for exploratory analysis. Whether specific variables (i.e., nodes) will be connected and the direction of the relationships do not need to be specified a priori when estimating a network model. This is useful when a strong theory does not exist, and the goal is to, instead, identify robust phenomena (e.g., conditional associations between explanatory factors). As argued in the introduction, there are many different factors that may influence the acceptance of large-scale solar, and the mechanism by which these factors influence acceptance are often unclear.

A good example of an unclear causal mechanism is the association between trust, the perceived effects of an energy technology and acceptance. Some research argue that trust causes differences in risk/benefit perception and, in turn, causes acceptance [34]. Other work suggests that the direction runs the other way: beliefs about impacts and benefits causes changes in trust and, in turn, acceptance [26,35]. Other explanatory factors like fairness and relationship quality are also hypothesised to influence acceptance through different mechanisms depending on the theory [20,26,35]. As such, a modelling approach that relaxes the assumption of causality and specific pathways is useful in this context and can assist future theory development and hypothesis generation.

The second reason for using an undirected network analysis approach is that these models do not assume acyclicity. That is, undirected network models do not assume that causality only flows in one direction (i.e., there are no feedback loops in the model). Not assuming acyclicity is useful in this context as many of the associations between psychological factors may be caused by bidirectional effects, or by unobserved (latent) variables. For example, perceived impacts may covary with perceived benefits due to independent causal effects that each variable has on the other, or due to an unobserved latent variable (e.g., an implicit attitude towards solar technology; [40,71]). Using a modelling approach that does not assume acyclicity, and therefore not ruling out feedback loops and the presence of latent variables, makes it easier to identify such effects to refine existing theories and inform the design of future confirmatory research.

Undirected network models also offer a unique perspective of practical importance by allowing variables to exert simultaneous influence on multiple others. Most statistical models applied in the context of understanding social acceptance focus on a singular dependent variable and thus obscure the intricate interconnections among independent variables. This approach proves especially advantageous when exploring complex acceptance phenomena, as variables can propagate effects through a web of intricate associations that evade clear visualization via conventional methods like linear regression. By modelling variable associations in this manner, it is possible to ascertain which variables have the greatest influence across the entire network. These insights carry significant practical implications for policy and decision-making, as it is possible to determine the most influential variables within a system, rather than fixating on single bivariate associations.

Gaussian Graphical Models (or partial correlation networks) were estimated using the 'bootnet' [72] package in RStudio. A Gaussian Graphical Model is a type of undirected network model that assumes a multivariate normal distribution. In its simplest form (without regularisation or pruning of edges), a Gaussian Graphical Model represents bivariate partial correlation coefficients between all pairs of variables included in the model, conditioning on all other variables. In this study, network models were estimated using the mixed graphical model [73] function inside bootnet, which uses nodewise regressions and the lasso regularisation penalty to reduce spurious edges (i.e., weak coefficients). The nodewise estimation approach was chosen so that the explained variability in sample (i.e., R^2) could be calculated for each node in the network.

Graphical representation of the results of the undirected network models were created using the *qgraph* package [74]. When plotted, the nodes in graphs represent each variable included in the analysis, and the links or 'edges' (associations been variables) can be interpreted as partial correlations or scaled regression coefficients (see Ref. [75] for a summary of how partial correlations correspond to regression coefficients). The thickness of the edges represents the strength of the regularised regression coefficient between two variables (nodes) after controlling for all other variables in the network. In other words, a thicker edges represent higher regression coefficients. Another way to interpret the network graphs is to think of each node as a distinct dependent (outcome) variable in a multiple linear regression model with all other nodes being independent (predictor) variables. The *spring* layout was specified to create the plot, which uses the Fruchterman-reingold algorithm [76] to determine spacing between nodes in a way that reflects the relative strength of associations [74]. Put simply, nodes that have stronger associations are placed closer together in the graph. The colour saturation and width of edges are scaled with reference to the strength of the coefficients.

3. Results

3.1. Correlations

Before estimating the undirected network model, unviariate correlation analyses indicated that explanatory variable are significantly associated with social acceptance to varying degrees. Feelings about the solar farm, perceived wider benefits, perceived distributional fairness and support for the renewable energy transition narrative all correlated strongly with social acceptance for the solar farm ($r > 0.65$).

3.2. Solar acceptance network model

Fig. 1 shows the results of the first undirected network model using all of the variables shown in Table 2. The selection of explanatory variables to include in the model was primarily informed by commonly applied theoretical models explaining community acceptance of energy-related projects [26,35]. Although there are some differences between these theoretical models regarding the mechanisms that lead to greater community acceptance, they all specify the importance of perceived impacts, perceived benefits, distributional fairness, procedural fairness, governance, trust, and relationship quality.

The whole sample was used to estimate the network model. In Fig. 1, the amount of shaded area around each node represents the amount of variability in that node that is explained by all the other nodes in the network (i.e., R^2). As shown in Fig. 1, roughly 75 % of the variability in the social acceptance of a local solar farm can be explained by the other variables in the model. The core results of the analysis indicate that societal benefits, distributional fairness, and support for the energy transition narrative, all have strong direct associations with acceptance. General support for renewables is directly linked to positive feelings, while concerns about the negative impacts of solar farms indirectly affects acceptance through reported feelings. Distribution fairness and support for renewables predict acceptance directly and indirectly through perceived societal benefits. Beliefs about fossil fuel exports and 'peaking' gas have more distant, indirect associations with acceptance, primarily via their correlations with worldviews and the broader support for renewables. Social factors like trust in the industry, governance, relationship quality, and procedural fairness, while having limited direct connections with acceptance, are indirectly associated via other variables.

The degree to which one node effects other nodes in the network was examined by calculating a strength measure for each node using the *qgraph* package [74]. Node strength is calculated by the summing the edge weights (i.e., regression coefficients) that each node shares with directly connected nodes. As shown in Fig. 1, perceived societal benefits has several direct and relatively strong connections with other nodes, which is why it has high node strength (as shown in Fig. 2). Conversely, perceived impacts has relatively few direct associations, which explains why it is a low strength node.

Table 2
Explanatory variable means, standard deviations and the correlation coefficients showing explanatory variables are significantly associated with social acceptance of a local solar farm development.

Variable	*M*	*SD*	Coefficient [95 % C.I]
Procedural fairness	4.67	1.22	.57** [.54, .59]
Trust in industry	4.62	1.29	.59** [.57, .61]
Trust in governance	4.74	1.23	.61** [.58, .63]
Relationship quality	4.73	1.17	.52** [.50, .55]
Distributive fairness	5.05	1.15	.70** [.68, .71]
Perceived local benefits	4.96	1.15	.62** [.59, .64]
Perceived societal benefits	5.16	1.14	.73** [.71, .74]
Perceived impacts	4.16	1.31	−.37** [-.40, −.33]
Role for renewables	5.07	1.10	.67** [.65, .69]
Role for gas	4.68	1.20	.22** [.19, .26]
Role for fossil fuel export	4.29	1.44	.20** [.16, .23]
Willingness to make trade offs	2.91	1.42	.15** [.12, .19]
Fragile	6.02	1.79	.38** [.35, .41]
Manageable	5.13	1.62	.30** [.26, .33]
Adaptable	4.14	1.67	.09** [.05, .12]
Unpredictable	4.53	1.78	−.12** [-.16, −.09]
Feelings	4.85	1.14	.82** [.81, .83]

Note. M and *SD* are used to represent mean and standard deviation, respectively. ** indicates $p < .01$.

The strength of each node suggests how easy it may be to change a given belief and the extent to which changing that belief would influence other associated beliefs. As perceived societal benefits is a high strength node, changing it may have a relatively strong effect on other nodes in the network. However, higher strength nodes may also be more difficult to change due to their connections with other nodes [62]. Put simply, a high strength node (in this case, a highly influential belief) may be more resistant to change due to it being 'held in place' by its associations with other nodes. It may, for example, be difficult to change perceived societal benefits without also changing beliefs about the energy transition narrative, distributional fairness and local benefits as these beliefs may have causal effects on perceived societal benefits (e.g.,

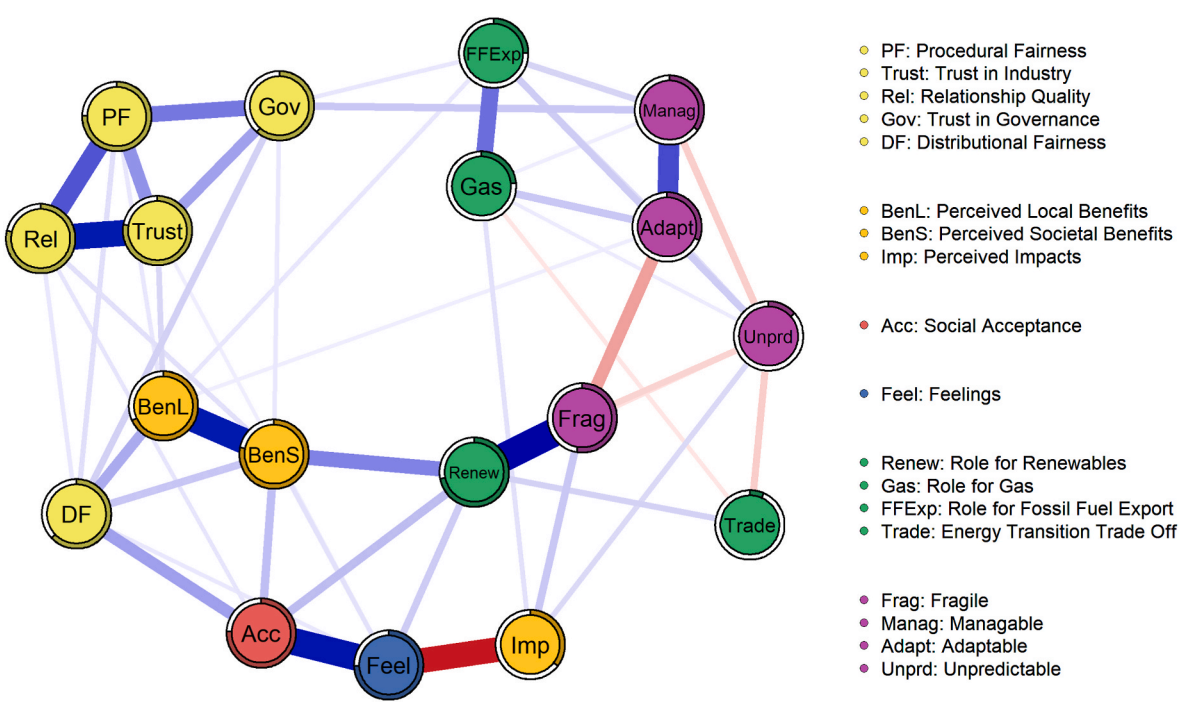

Fig. 1. Undirected network model of local solar farm acceptance showing that perceived societal benefits, distributional fairness, and general support for renewables, all have strong direct associations with local acceptance of a large-scale solar farm.

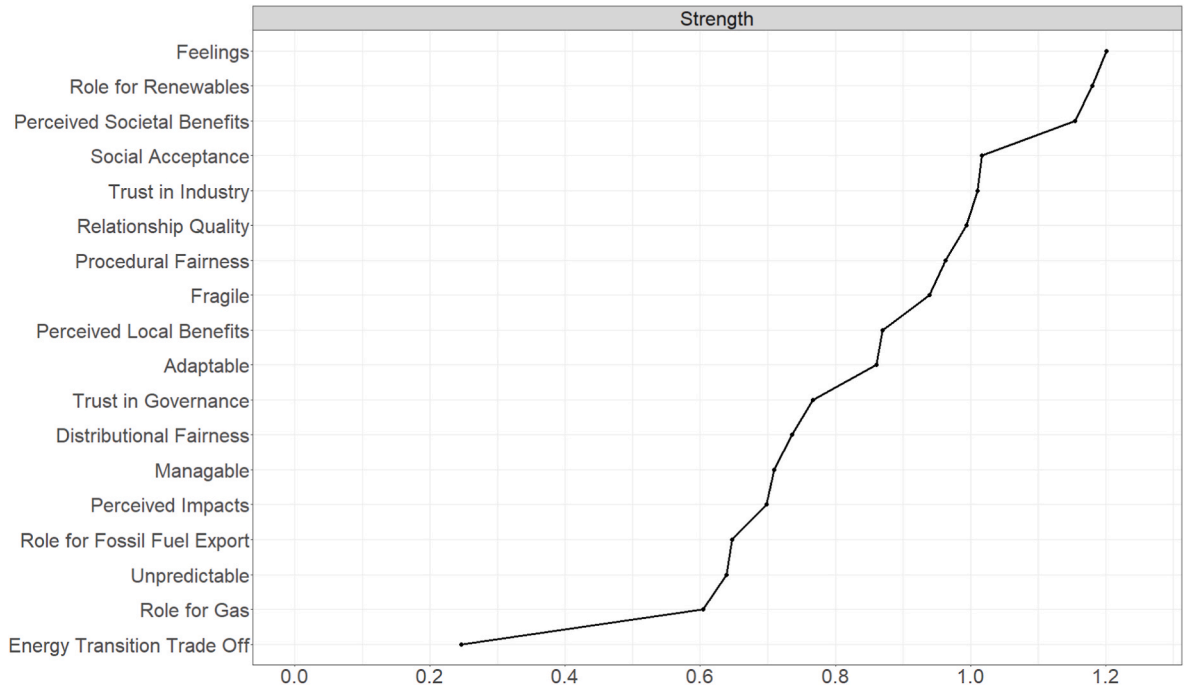

Fig. 2. Plot showing the strength of each node in the social licence network ordered from highest to lowest strength. The figure shows that the 'feelings' node had the strongest influence on all other nodes in the social licence network.

I support the energy transition narrative, therefore I think solar farms are beneficial for society). Conversely, perceived impacts is a relatively low strength node but still has a relative strong indirect connection with acceptance (via feelings). This may mean that changing beliefs about the impacts of solar farms may be easier as it is less dependent on other beliefs in the network. However, it is important to note that perceived impacts may be influenced by other beliefs that were not included in the network.

3.3. Unpacking the narrative

As beliefs about the energy transition seemed to be associated with worldviews in the first network model, an additional network model was estimated to explore the conditional associations between each belief item and worldviews. Fig. 3 shows a graph representing another undirected network model estimated in the same way as the previous network model. The whole sample again was used to estimate the network model. In the second graph, though, the colour saturation and width of edges are scaled with reference to the strongest coefficient of .73. As seen in Fig. 3, the worldview items cluster in the middle of the graph and tend to have the strongest conditional associations with beliefs about renewables (orange-coloured nodes). Like the first network model, *role for gas, role for fossil fuel exports* and *willingness to make trade off* items were quite isolated in the graph and had weak or no conditional associations with worldviews.

Fig. 4 shows the relative strength of each node in the energy transition narrative network. As seen in Fig. 4, the belief that Australia should transition faster to using renewables was the strongest belief in

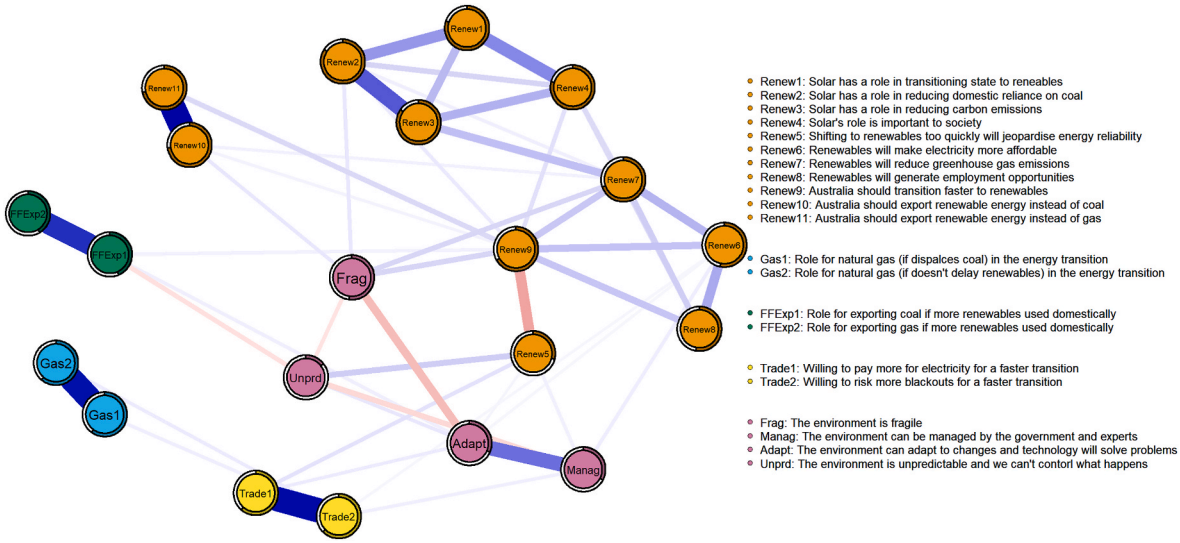

Fig. 3. Network graph showing the conditional associations between the beliefs about Australia's energy transition and worldviews. The graph shows direct relationships between energy transition beliefs and broader environmental worldviews.

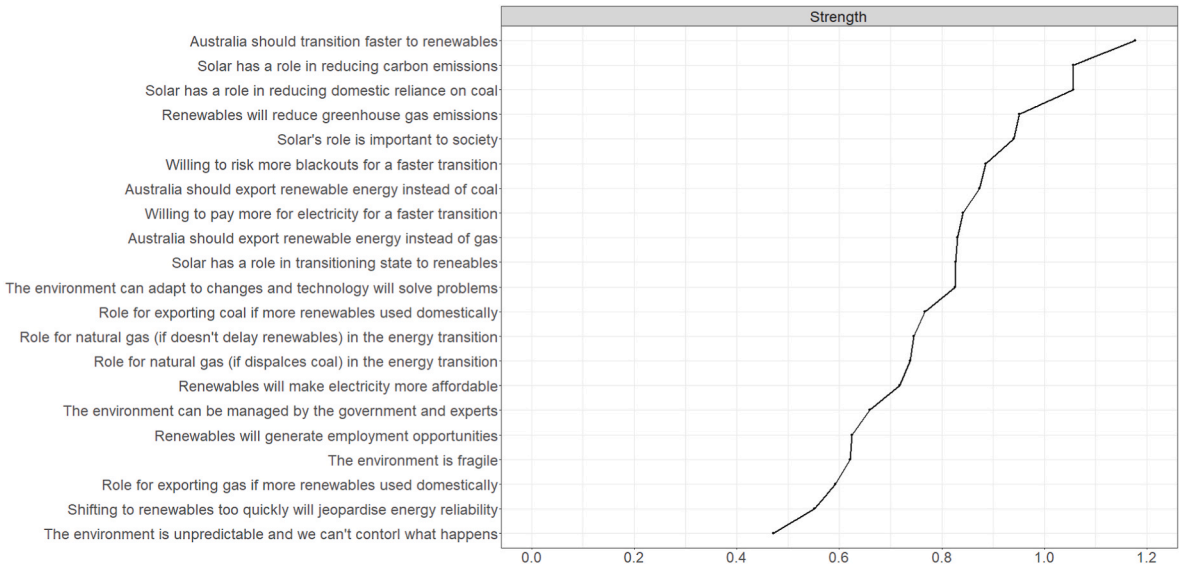

Fig. 4. Plot showing the strength of each node in the energy transition narrative network ordered from highest to lowest strength. The figure shows that believing Australia should transition faster to using renewables had the strongest influence on other beliefs in the network.

the network. As such, changing people's level of agreement with this belief should have the greatest influence on all other beliefs in the network. Conversely, believing that the environment is unpredictable (a worldview item) had the weakest overall effect on all other beliefs in the network.

4. Discussion

This research explored the socio-psychological factors that influence local solar farm acceptance using a network modelling approach. In accordance with past social acceptance research, perceived benefits, feelings and distributional fairness were all directly associated with local solar farm acceptance [20,24,26]. However, unlike similar research in other energy domains, other social factors (e.g., trust and relationship quality) did not have direct associations with acceptance and had relatively weak indirect effects via typical mediating variables (e.g., perceived benefits and impacts). The findings also suggest that support for renewable energy in general (i.e. the energy transition narrative) is an important factor helping to explain local solar farm acceptance. This section will expand on these main empirical findings and outline some directions for future research with reference to relevant theory. The discussion will end by highlighting some of the study's limitations and summarising the main implications.

4.1. Broader beliefs about the energy transition

One novel finding in this research was the relatively strong association between local solar farm acceptance and general support for renewable energy. Support for renewables had one of the strongest bivariate correlations with acceptance and was directly related to both acceptance and positive feelings when controlling for all other variables in the first network model. One explanation for this finding is that people tend to think about solar farms, and their benefits, in a broader, more abstract way. Put another way, it seems that local solar farm acceptance is strongly influenced by a positive attitude towards renewables in general compared to a more considered assessment of the local impacts (both positive and negative) of a solar farm. This explanation is supported by past research showing that people tend to think about renewable energy technologies at an abstract level, which tends to result in a positive attitude [4,9]. As other research highlights, a positive attitude predicts higher levels of perceived benefits and lower levels of perceived risk/cost [40,41].

The strong link between the 'fragile' worldview and support for renewables also highlights the importance of a broader, and in this case more fundamental, attitude. It was found that believing that the environment is fragile, and will only be protected if there are large changes in human behaviour, was strongly associated with support for renewables in the network models. Although conditional associations in the network do not infer causal effects, theory would suggest that the 'fragile' worldview has a causal effect on other factors as it is a more fundamental attitude. Theories that explicate the role of 'worldviews' and/or 'values' posit that they influence energy technology acceptance via their effect on more contextual beliefs the (e.g., perceived risk and benefits of a technology; [29,46]). The findings, overall, suggest that those who endorse this 'fragile' worldview tend to be more accepting of local solar farms. It is, however, yet to be seen how strong this effect remains when people have more exposure to local effects of solar farm developments.

The second network model (Fig. 3) reveals how the 'fragile' worldview relates with specific beliefs about the energy transition. Fig. 3 shows that thinking the environment is fragile is most strongly associated with two specific beliefs about the role of renewables. Namely, the belief that Australia should be transitioning faster and the belief that renewables will reduce greenhouse gas emissions. As shown in Fig. 4, these two beliefs have some of the strongest effects on all other nodes in the network. The finding suggests that these two beliefs, which seem to be shaped by a specific worldview, have a strong influence on people's overall attitude toward renewables and the energy transition more broadly. Other beliefs about the potential direct effects of using more renewables (e.g., energy reliability, energy cost and employment opportunities) have a relatively weak influence on the network. These findings, again, support the idea that how people think about renewables in the abstract primarily shapes their overall attitude towards them.

Although some broader beliefs about the renewable energy transition seem to help explain local acceptance, others seem less important. Another interesting finding was that willingness to make trade-offs was quite isolated in the network models. This is a surprising result as strong support for renewables should be reflected in an increased willingness to make personal sacrifices if required (i.e., pay more for electricity or experience more blackouts). This finding suggests that while people tend to have a positive attitude towards renewables, such an attitude may not necessarily translate into behaviour that will facilitate a faster renewable energy transition (e.g., willingness to pay more for energy if

necessary). Future research is needed to test whether increases in price, or more unreliable energy supply, leads to lower levels of support for renewables. Such research would have significant implications for energy policy.

Beliefs about the role of gas and fossil fuel exports were also quite distinct in the network models. This finding suggests that beliefs about the role for renewables are uninfluenced by beliefs about other components of Australia's current energy system. For example, believing that there is a role for gas (if it does not delay renewables) was unrelated to expressed support for renewables. This is another surprising outcome as people with pro-environmental beliefs tend be more supportive of renewables, but less supportive of fossil fuels [5]. This may be due to the ambivalent role of gas as both a fossil fuel and a 'peaking' energy source which can support a transition to renewables. The findings also suggest that if the continued use of fossil fuels does not impede the development of renewable energy technologies, views about the role of fossil fuels will have a limited effect on attitudes towards renewables and, in turn, local solar farms.

4.2. Perceived impacts and benefits

Further supporting the idea that abstract considerations shape attitudes towards solar farms was the direct link between perceived societal benefits and local acceptance. The results suggest that support for renewables in general informs people's views about the societal benefits of solar farms, which in turn influence local acceptance. However, perceived local benefits, while associated with societal benefits and distributional fairness, had no direct link with local solar farm acceptance. It is likely that those who agree that solar farms will bring about societal benefits also tend to agree that they will bring about local benefits even though this has no direct effect on acceptance. Again, it seems that more abstract views about the wider benefits of renewables and solar farms are the primary beliefs informing people's expressed support for a local solar farm.

The strong association between feelings, impacts and acceptance also warrants further investigation. While distributional fairness, role for renewables and social benefits all have direct associations with both acceptance and feelings, impacts only seems to associate with acceptance via feelings (i.e., no direct effect). As perceived impacts is quite isolated from the other direct predictors of acceptance, it is possible that people's attitude towards solar farms informed their expressed concern about the local impacts (i.e., the causality runs the other way). Put another way, if people like the idea of using renewables they will have a positive attitude towards solar farms and, therefore, anticipate that the impacts will be low. The causal relationship is also supported by the theory behind the affect heuristic in that the liking or disliking of an object informs one's explicit beliefs about that object [40,41]. For example, any feelings associated from negative imagery may directly affect perceived impacts and social acceptance of solar farms.

4.3. Other contextual factors

While many other contextual factors were included in the network model, only perceived *distributional fairness* was directly associated with local solar farm acceptance. As distributional fairness was associated with both perceived benefit factors (local and societal) and not associated with perceived impacts, it may be the case that people thought more about the benefits of a solar farm in general when answering the distributional fairness questions as opposed to thinking about the local *process* of fairly distributing impacts and benefits. Regardless, the results suggest that believing the benefits of a solar farm will be distributed fairly in a region is strongly associated with local acceptance directly and indirectly via perceived local and societal benefits.

It was surprising that other contextual factors like trust in the industry and relationship quality were quite isolated in the network model. It is, though, difficult to claim that these contextual factors will not be important for developing local support for solar farms. Past research tends to find that the contextual factors measured in this study help to explain local acceptance of many different energy-related projects [24,26,35]. However, the studies that have found these effects tend to be conducted in regions where people have experience living near energy-related projects [26,35]. Therefore, the participants sampled in these previous studies had exposure to the social actors involved and have, therefore, had time to develop attitudes towards them. As most Australians have not experienced living near a solar farm, most would not have strong beliefs about the social actors involved or the governance practices. Overall, the results suggest that people are not yet considering the social context in which solar farms will be built and operated and, instead, tend to rely more on their general attitude towards renewables to inform their reported level of local acceptance.

The findings in this research should be contextualised in the broader setting of Australia's energy transition progress. As there is a limited number of people currently living near solar farms in Australia, participants in this study were asked to share their views about living near a hypothetical utility-scale solar farm. There are known limitations to eliciting realistic responses to hypothetical scenarios [77] and not having real-world experience living near a solar farm may have made it difficult for respondents to consider the impact of contextual factors. An inability to fully understand and imagine local impacts may also explain why support for renewables seemed to have a stronger influence in the network models than the perceived local effects of solar farms. Most people, without having first-hand experience, may not know what the consequences of solar farms are and how they would affect their lives. In the future, research using hypothetical scenarios to elicit responses may consider alternative methods for supporting the validity of the scenarios. A randomized controlled study could be used to verify changes in research outcomes. In such a study, the treatment group could receive visual cues (e.g., pictures, videos) to help them conceptualize the physical impact of large-scale solar farms or other renewable energy infrastructure.

4.4. Limitations and future research

This study's foremost limitation stems from its reliance on a hypothetical situation. Consequently, it is likely that broader attitudes and views played a more prominent role in this research, given that many participants lacked firsthand experience with the social context, benefits and costs of living near a large-scale solar farm. This caveat should be considered when extrapolating the results to real-world scenarios, particularly in the context of formulating policy recommendations regarding the importance of trust and social factors. As a potential avenue for future research, a longitudinal assessment of acceptance and attitudes during the progressive rollout of large-scale infrastructure could provide valuable insights. Building on past research, it is expected that perceived trust, local benefits and local impacts would become more influential over time. However, an intriguing question to explore is the extent to which broader energy transition narratives continue to influence acceptance or undergo changes due to experience living near a development.

While people in developed countries tend to support a renewable energy transition [78], it is not yet clear how supportive they are of different ways of implementing renewable energy technologies. It seems that most people, at this stage, are not thinking about potential compromises they may need in a societal transition to using more renewables. As trade-offs may be necessary in the energy transition, future research should consider how receptive people would be to different policies that may cause different types of trade-offs (both at a community level and a societal level). Such research would help to inform future energy policy to ensure that large-scale renewable energy projects are developed in socially sustainable way for host communities and wider society.

One limitation of the analysis approach used in this research is that

undirected network models do not represent directional causal effects. This limitation carries relevance for policy implications, as it cautions against making assumptions that certain factors, such as acceptance, do not exert their own causal influences on other variables within the network. It is plausible, for instance, that implicit attitudes towards solar farms, characterized by a general liking or disliking, may precede and influence reported acceptance, subsequently shaping individuals' evaluations of the associated impacts and benefits. To address this limitation, future research could explore the application of directed network models, such as Bayesian networks [79], within a causal inference framework [80]. Directed graphs, when informed by causal assumptions, can shed light on the direction and strength of causal effects, unlike undirected models. The trade-off is that directed graphs depend on stronger causal assumptions.

Additionally, it is worth noting that network models, both directed and undirected, can be effectively estimated with longitudinal or panel data [73]. This opens opportunities for research to track how beliefs and attitudes evolve over time, especially when confronted with tangible changes on the ground, such as shifts in employment, economic conditions, or the local environment. Furthermore, experimental designs represent another avenue of investigation, as it is possible to manipulate specific perceptual factors, like perceived benefits, to discern their causal effects on other variables.

5. Conclusion

This study investigated the extent to which support for energy transition narratives influence local acceptance of solar farms and other social licence factors. This is, to the authors' knowledge, the first time that a network modelling approach has been used to examine how social licence factors influence local acceptance of large-scale renewable energy infrastructure. The findings of the study indicate that: (1) beliefs about the environment are connected to views on the use of renewable energy; (2) broader beliefs about the role of renewable energy sources impact local acceptance of solar farms; and (3) beliefs about the role of fossil fuels are only weakly linked to local acceptance. Overall, the findings suggest that at this early stage of the renewable energy transition, people's expressed level of support for local large-scale energy projects is primarily informed by more abstract attitudes and beliefs about the environment and the utility of renewable energy technology. At this stage, expressed support for local projects is less dependent on local factors such trust, procedural fairness and governance.

It is, however, likely that most people have not yet formed their beliefs about the social context in which solar farms operate. This means that support for local solar farms may start to change as residents gain more experience living near utility-scale renewables and the associated infrastructure or if negative feelings are invoked through negative images of solar farms (e.g., solar farms in disrepair). It is possible that if a negative sentiment takes hold at local levels, broader public narratives about the energy transition may become more negative, putting the renewable energy industry at risk.

It is clear from this research that although local factors are important (e.g., perceived local benefits and impacts) the broader narrative about the role for renewables also needs consideration. It is, therefore, important that people of influence (e.g., politicians, media, relevant industries) understand they how they shape public narratives about the role for renewables can impact community responses to proposed renewable projects. Although this research was conducted with an Australian sample, this finding has implications for culturally and contextually similar populations that are currently transitioning from fossil fuels to renewable energy technologies. These findings underscore the global relevance of shaping narratives around renewables during the transition from fossil fuels to clean energy technologies.

Given the importance of accelerating the decarbonization of the world's energy systems [81], any delays, slow-downs, or fractured changes in the transition to renewables could hinder global environmental and economic progress. Society is relying on new technology and its uptake to relieve the pressures of a changing climate, pollution, geopolitical tensions and war-induced economic instability [82–85]. This research highlights that wider public narratives, and how they shape social licence factors, are important components of a successful energy transition. Continuing to monitor local acceptance and understanding the psychosocial factors that facilitate or impede it is essential to ensuring that countries can continue their energy transitions to reach net-zero goals.

Credit author

Scovell: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. McCrea: Conceptualization, Methodology, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Funding acquisition. Walton: Conceptualization, Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing, Funding acquisition. Poruschi: Conceptualization, Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

The authors acknowledge funding for the authors time and the data collection was provided by the Commonwealth Scientific and Industrial Research Organisation (CSIRO). The data collection was carried out in 2020. The views expressed here do not reflect the views of the Commonwealth Government of Australia nor those of the CSIRO.

References

[1] Victoria M, Haegel N, Peters IM, Sinton R, Jäger-Waldau A, del Cañizo C, et al. Solar photovoltaics is ready to power a sustainable future. Joule 2021;5:1041–56. https://doi.org/10.1016/J.JOULE.2021.03.005.
[2] Carlisle JE, Kane SL, Solan D, Joe JC. Support for solar energy: examining sense of place and utility-scale development in California. Energy Res Social Sci 2014;3:124–30. https://doi.org/10.1016/J.ERSS.2014.07.006.
[3] Wüstenhagen R, Wolsink M, Bürer MJ. Social acceptance of renewable energy innovation: an introduction to the concept. Energy Pol 2007;35:2683–91. https://doi.org/10.1016/j.enpol.2006.12.001.
[4] Cousse J. Still in love with solar energy? Installation size, affect, and the social acceptance of renewable energy technologies. Renew Sustain Energy Rev 2021;145:111107. https://doi.org/10.1016/J.RSER.2021.111107.
[5] Hobman EV, Ashworth P. Public support for energy sources and related technologies: the impact of simple information provision. Energy Pol 2013;63:862–9. https://doi.org/10.1016/j.enpol.2013.09.011.
[6] Visschers VHM, Siegrist M. Find the differences and the similarities: relating perceived benefits, perceived costs and protected values to acceptance of five energy technologies. J Environ Psychol 2014;40:117–30. https://doi.org/10.1016/j.jenvp.2014.05.007.
[7] Volken S, Wong-Parodi G, Trutnevyte E. Public awareness and perception of environmental, health and safety risks to electricity generation: an explorative interview study in Switzerland. J Risk Res 2019;22:432–47. https://doi.org/10.1080/13669877.2017.1391320.
[8] Schumacher K, Krones F, McKenna R, Schultmann F. Public acceptance of renewable energies and energy autonomy: a comparative study in the French, German and Swiss Upper Rhine region. Energy Pol 2019;126:315–32. https://doi.org/10.1016/J.ENPOL.2018.11.032.
[9] Sütterlin B, Siegrist M. Public acceptance of renewable energy technologies from an abstract versus concrete perspective and the positive imagery of solar power. Energy Pol 2017;106:356–66. https://doi.org/10.1016/j.enpol.2017.03.061.

[10] Boudet HS. Public perceptions of and responses to new energy technologies. Nat Energy 2019;4:446–55. https://doi.org/10.1038/s41560-019-0399-x.
[11] Devine-Wright P. Rethinking NIMBYism: the role of place attachment and place identity in explaining place-protective action. J Community Appl Soc Psychol 2009;19:426–41. https://doi.org/10.1002/CASP.1004.
[12] Batel S, Devine-Wright P. A critical and empirical analysis of the national-local 'gap' in public responses to large-scale energy infrastructures. J Environ Plann Manag 2014;58:1076–95. https://doi.org/10.1080/09640568.2014.914020.
[13] Measham T, McCrea R, Poruschi L, Walton A, O'Sullivan D. The role of large-scale solar in transitioning to a low carbon energy system: social and economic issues and an emerging research agenda. CSIRO; 2021. https://doi.org/10.25919/ndrm-9v11.
[14] Walker C, Baxter J. Procedural justice in Canadian wind energy development: a comparison of community-based and technocratic siting processes. Energy Res Social Sci 2017;29:160–9. https://doi.org/10.1016/J.ERSS.2017.05.016.
[15] Devine-Wright P. Beyond NIMBYism: towards an integrated framework for understanding public perceptions of wind energy. Wind Energy 2005;8:125–39. https://doi.org/10.1002/WE.124.
[16] Bidwell D. The role of values in public beliefs and attitudes towards commercial wind energy. Energy Pol 2013;58:189–99. https://doi.org/10.1016/j.enpol.2013.03.010.
[17] Carlisle JE, Kane SL, Solan D, Bowman M, Joe JC. Public attitudes regarding large-scale solar energy development in the U.S. Renew Sustain Energy Rev 2015;48:835–47. https://doi.org/10.1016/J.RSER.2015.04.047.
[18] Carlisle JE, Solan D, Kane SL, Joe J. Utility-scale solar and public attitudes toward siting: a critical examination of proximity. Land Use Pol 2016;58:491–501. https://doi.org/10.1016/J.LANDUSEPOL.2016.08.006.
[19] Nilson RS, Stedman RC. Are big and small solar separate things?: the importance of scale in public support for solar energy development in upstate New York. Energy Res Social Sci 2022;86:102449. https://doi.org/10.1016/J.ERSS.2021.102449.
[20] Huijts NMA, Molin EJE, Steg L. Psychological factors influencing sustainable energy technology acceptance: a review-based comprehensive framework. Renew Sustain Energy Rev 2012;16:525–31. https://doi.org/10.1016/j.rser.2011.08.018.
[21] Scovell MD. Explaining hydrogen energy technology acceptance: a critical review. Int J Hydrogen Energy 2022;47:10441–59. https://doi.org/10.1016/j.ijhydene.2022.01.099.
[22] Gupta N, Fischer ARH, Frewer LJ. Socio-psychological determinants of public acceptance of technologies: a review. Publ Understand Sci 2012;21:782–95. https://doi.org/10.1177/0963662510392485.
[23] Bronfman NC, Jiménez RB, Arévalo PC, Cifuentes LA. Understanding social acceptance of electricity generation sources. Energy Pol 2012;46:246–52. https://doi.org/10.1016/J.ENPOL.2012.03.057.
[24] Huijts NMA, Molin EJE, van Wee B. Hydrogen fuel station acceptance: a structural equation model based on the technology acceptance framework. J Environ Psychol 2014;38:153–66. https://doi.org/10.1016/j.jenvp.2014.01.008.
[25] Terwel BW, Harinck F, Ellemers N, Daamen DDL. Competence-based and integrity-based trust as predictors of acceptance of carbon dioxide capture and storage (CCS). Risk Anal 2009;29:1129–40. https://doi.org/10.1111/j.1539-6924.2009.01256.x.
[26] Walton A, McCrea R. Understanding social licence to operate for onshore gas development: how the underlying drivers fit together. Appl Energy 2020;279:115750. https://doi.org/10.1016/J.APENERGY.2020.115750.
[27] Warren CR, Lumsden C, O'Dowd S, Birnie RV. 'Green on green': public perceptions of wind power in scotland and Ireland. J Environ Plann Manag 2011;48:853–75. https://doi.org/10.1080/09640560500294376.
[28] Wolsink M. Wind power and the NIMBY-myth: institutional capacity and the limited significance of public support. Renew Energy 2000;21:49–64. https://doi.org/10.1016/S0960-1481(99)00130-5.
[29] De Groot JIM, Steg L, Poortinga W. Values, perceived risks and benefits, and acceptability of nuclear energy. Risk Anal 2013;33:307–17. https://doi.org/10.1111/j.1539-6924.2012.01845.x.
[30] de Groot JIM, Schweiger E, Schubert I. Social influence, risk and benefit perceptions, and the acceptability of risky energy technologies: an explanatory model of nuclear power versus shale gas. Risk Anal 2020;40:1226–43. https://doi.org/10.1111/RISA.13457.
[31] L'Orange Seigo S, Dohle S, Siegrist M. Public perception of carbon capture and storage (CCS): a review. Renew Sustain Energy Rev 2014;38:848–63. https://doi.org/10.1016/j.rser.2014.07.017.
[32] Visschers VHM, Keller C, Siegrist M. Climate change benefits and energy supply benefits as determinants of acceptance of nuclear power stations: investigating an explanatory model. Energy Pol 2011;39:3621–9. https://doi.org/10.1016/j.enpol.2011.03.064.
[33] Visschers VHM, Siegrist M. Differences in risk perception between hazards and between individuals. In: Raue M, Lermer E, Streicher B, editors. Psychological perspectives on risk and risk analysis: theory, models, and applications. Cham: Springer; 2018. p. 63–80. https://doi.org/10.1007/978-3-319-92478-6_3.
[34] Siegrist M. The influence of trust and perceptions of risks and benefits on the acceptance of gene technology. Risk Anal 2000;20:195–204. https://doi.org/10.1111/0272-4332.202020.
[35] Moffat K, Zhang A. The paths to social licence to operate: an integrative model explaining community acceptance of mining. Resour Pol 2014;39:61–70. https://doi.org/10.1016/j.resourpol.2013.11.003.
[36] Midden CJH, Huijts NMA. The role of trust in the affective evaluation of novel risks: the case of CO2 storage. Risk Anal 2009;29:743–51. https://doi.org/10.1111/j.1539-6924.2009.01201.x.
[37] Siegrist M, Cousin ME, Kastenholz H, Wiek A. Public acceptance of nanotechnology foods and food packaging: the influence of affect and trust. Appetite 2007;49:459–66. https://doi.org/10.1016/j.appet.2007.03.002.
[38] Zajonc RB. Feeling and thinking: preferences need no inferences. Am Psychol 1980;35:151–75. https://doi.org/10.1037/0003-066X.35.2.151.
[39] Clore GL, Schnall S. The influence of affect on attitude. In: Albarracin Dolores, Johnson Blair T, Zanna Mark P, editors. Handbook of attitudes. New York: Psychology Press; 2005. p. 437–89.
[40] Finucane ML, Alhakami A, Slovic P, Johnson SM. The affect heuristic in judgments of risks and benefits. J Behav Decis Making 2000;13:1–17. https://doi.org/10.1002/(SICI)1099-0771(200001/03)13:1<1::AID-BDM333>3.0.CO;2-S.
[41] Slovic P, Finucane ML, Peters E, MacGregor DG. The affect heuristic. Eur J Oper Res 2007;177:1333–52. https://doi.org/10.1016/j.ejor.2005.04.006.
[42] Perlaviciute G, Steg L. Contextual and psychological factors shaping evaluations and acceptability of energy alternatives: integrated review and research agenda. Renew Sustain Energy Rev 2014;35:361–81. https://doi.org/10.1016/j.rser.2014.04.003.
[43] Corner A, Venables D, Spence A, Poortinga W, Demski C, Pidgeon N. Nuclear power, climate change and energy security: exploring British public attitudes. Energy Pol 2011;39:4823–33. https://doi.org/10.1016/j.enpol.2011.06.037.
[44] Perlaviciute G, Steg L, Contzen N, Roeser S, Huijts N. Emotional responses to energy projects: insights for responsible decision making in a sustainable energy transition. Sustainability 2018;10. https://doi.org/10.3390/su10072526.
[45] Spence A, Poortinga W, Pidgeon N, Lorenzoni I. Public perceptions of energy choices: the influence of beliefs about climate change and the environment. Energy Environ 2010;21:385–407. https://doi.org/10.1260/0958-305X.21.5.385.
[46] Whitfield SC, Rosa EA, Dan A, Dietz T. The future of nuclear power: value orientations and risk perception. Risk Anal 2009;29:425–37. https://doi.org/10.1111/j.1539-6924.2008.01155.x.
[47] van der Linden S. The social-psychological determinants of climate change risk perceptions: towards a comprehensive model. J Environ Psychol 2015;41:112–24. https://doi.org/10.1016/j.jenvp.2014.11.012.
[48] Douglas M, Wildavsky A. How can we know the risks we face? Why risk selection is a social Process1. Risk Anal 1982;2:49–58. https://doi.org/10.1111/j.1539-6924.1982.tb01365.x.
[49] McCrea R, Leviston Z, Walker IA. Climate change skepticism and voting behavior: what causes what? Environ Behav 2016;48:1309–34. https://doi.org/10.1177/0013916515599571.
[50] Boutilier RG. Narratives and networks model of the social licence. Resour Pol 2020;69:101869. https://doi.org/10.1016/j.resourpol.2020.101869.
[51] Devine-Wright P, Batel S, Aas O, Sovacool B, LaBelle MC, Ruud A. A conceptual framework for understanding the social acceptance of energy infrastructure: insights from energy storage. Energy Pol 2017;107:27–31. https://doi.org/10.1016/j.enpol.2017.04.020.
[52] Hermwille L. The role of narratives in socio-technical transitions—fukushima and the energy regimes of Japan, Germany, and the United Kingdom. Energy Res Social Sci 2016;11:237–46. https://doi.org/10.1016/j.erss.2015.11.001.
[53] Hazboun SO, Boudet HS. Natural gas–friend or foe of the environment? Evaluating the framing contest over natural gas through a public opinion survey in the Pacific Northwest. Environmental Sociology 2021;7:368–81. https://doi.org/10.1080/23251042.2021.1904535.
[54] Heffron RJ, McCauley D, Sovacool BK. Resolving society's energy trilemma through the Energy Justice Metric. Energy Pol 2015;87:168–76. https://doi.org/10.1016/j.enpol.2015.08.033.
[55] Australian Bureau of Statistics. Australian national accounts: national income, expenditure and product. June 2022 | Australian Bureau of Statistics; 2022. https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-national-income-expenditure-and-product/latest-release. [Accessed 1 December 2022].
[56] World Bank. World integrated trade solution (WITS) | data on export, import, tariff, NTM. 2022. https://wits.worldbank.org/. [Accessed 1 December 2022].
[57] Department of Industry Science and Resources. Resources and energy quarterly, https://www.industry.gov.au/publications/resources-and-energy-quarterly. [Accessed 1 December 2022].
[58] International Energy Agency. World energy outlook. 2022. https://www.iea.org/reports/world-energy-outlook-2022. [Accessed 1 December 2022].
[59] Walton A, Mccrea R, Poruschi L, Measham T, Gillespie M, O'Sullivan D. Australian attitudes and perceptions of large-scale solar: social licence and the role of solar in the energy transition. CSIRO; 2021. 10.25919%2Ffqbk-0y13.
[60] Finstad K. Response interpolation and scale sensitivity: evidence against 5-point scales, vol. 5; 2009.
[61] Steg L, Sievers I. Cultural theory and individual perceptions of environmental risks. Environ Behav 2000;32:250–69. https://doi.org/10.1177/00139160021972513.
[62] Dalege J, Borsboom D, Van Harreveld F, Van den Berg H, Conner M, Van der Maas HLJ. Toward a formalized account of attitudes: the causal attitude network (CAN) model. Psychol Rev 2016;123:2–22. https://doi.org/10.1037/A0039802.
[63] Borsboom D, Cramer AOJ. Network analysis: an integrative approach to the structure of psychopathology. Annu Rev Clin Psychol 2013;9:91–121. https://doi.org/10.1146/ANNUREV-CLINPSY-050212-185608.
[64] Costantini G, Epskamp S, Borsboom D, Perugini M, Mõttus R, Waldorp LJ, et al. State of the aRt personality research: a tutorial on network analysis of personality data in R. J Res Pers 2015;54:13–29. https://doi.org/10.1016/J.JRP.2014.07.003.
[65] Van Der Maas HLJ, Dolan CV, Grasman RPPP, Wicherts JM, Huizenga HM, Raijmakers MEJ. A dynamical model of general intelligence: the positive manifold of intelligence by mutualism. Psychol Rev 2006;113:842. https://doi.org/10.1037/0033-295X.113.4.842.

[66] Zwicker MV, Nohlen HU, Dalege J, Gruter GJM, van Harreveld F. Applying an attitude network approach to consumer behaviour towards plastic. J Environ Psychol 2020;69:101433. https://doi.org/10.1016/J.JENVP.2020.101433.
[67] Langley DJ, Wijn R, Epskamp S, van Bork R. Encouraging vaccination behavior through online social media. In: D'Ascenzo F, Magni M, Lazazzara A, Za S, editors. Lecture notes in information systems and organisation, vol. 19. Cham: Springer; 2016. p. 307–18.
[68] Hevey D. Network analysis: a brief overview and tutorial. Health Psychology and Behavioral Medicine 2018;6:301–28. https://doi.org/10.1080/21642850.2018.1521283.
[69] Bhushan N, Mohnert F, Sloot D, Jans L, Albers C, Steg L. Using a Gaussian graphical model to explore relationships between items and variables in environmental psychology research. Front Psychol 2019;10:1050. https://doi.org/10.3389/FPSYG.2019.01050/BIBTEX.
[70] Chambon M, Dalege J, Elberse JE, van Harreveld F. A psychological network approach to attitudes and preventive behaviors during pandemics: a COVID-19 study in the United Kingdom and The Netherlands: social psychological and personality. Science 2021;13:233–45. https://doi.org/10.1177/19485506211002420.
[71] Scott SE, Inbar Y, Wirz CD, Brossard D, Rozin P. An overview of attitudes toward genetically engineered food. Annu Rev Nutr 2018;38:459–79. https://doi.org/10.1146/ANNUREV-NUTR-071715-051223.
[72] Epskamp S, Borsboom D, Fried EI. Estimating psychological networks and their accuracy: a tutorial paper. Behav Res 2018;50:195–212. https://doi.org/10.3758/s13428-017-0862-1.
[73] Haslbeck JMB, Waldorp LJ. MGM: estimating time-varying mixed graphical models in high-dimensional data. J Stat Software 2020;93. https://doi.org/10.18637/JSS.V093.I08.
[74] Epskamp S, Cramer AOJ, Waldorp LJ, Schmittmann VD, Borsboom D. qgraph: network visualizations of relationships in psychometric data. J Stat Software 2012;48:1–18.
[75] Epskamp S, Waldorp LJ, Mõttus R, Borsboom D. The Gaussian graphical model in cross-sectional and time-series data. Multivariate Behav Res 2018;53:453–80. https://doi.org/10.1080/00273171.2018.1454823.
[76] Fruchterman TMJ, Reingold EM. Graph drawing by force-directed placement. Software Pract Ex 1991;21:1129–64. https://doi.org/10.1002/spe.4380211102.
[77] Ajzen I, Brown TC, Carvajal F. Explaining the discrepancy between intentions and actions: the case of hypothetical bias in contingent valuation. Pers Soc Psychol Bull 2004;30:1108–21. https://doi.org/10.1177/0146167204264079.
[78] Stokes LC, Warshaw C. Renewable energy policy design and framing influence public support in the United States. Nat Energy 2017;2:1–6. https://doi.org/10.1038/nenergy.2017.107.
[79] McNally RJ, Mair P, Mugno BL, Riemann BC. Co-morbid obsessive–compulsive disorder and depression: a Bayesian network approach. Psychol Med 2017;47:1204–14. https://doi.org/10.1017/S0033291716003287.
[80] Rohrer JM. Thinking clearly about correlations and causation: graphical causal models for observational data. Advances in Methods and Practices in Psychological Science 2018;1:27–42. https://doi.org/10.1177/2515245917745629.
[81] Geels FW, Sovacool BK, Schwanen T, Sorrell S. Sociotechnical transitions for deep decarbonization. Science 2017;357:1242–4. https://doi.org/10.1126/science.aao3760.
[82] Dincer I. Renewable energy and sustainable development: a crucial review. Renew Sustain Energy Rev 2000;4:157–75. https://doi.org/10.1016/S1364-0321(99)00011-8.
[83] Landrigan PJ, Fuller R, Acosta NJR, Adeyi O, Arnold R, Basu N, et al. The Lancet Commission on pollution and health. Lancet 2018;391:462–512. https://doi.org/10.1016/S0140-6736(17)32345-0.
[84] Panwar NL, Kaushik SC, Kothari S. Role of renewable energy sources in environmental protection: a review. Renew Sustain Energy Rev 2011;15:1513–24. https://doi.org/10.1016/j.rser.2010.11.037.
[85] Vakulchuk R, Overland I, Scholten D. Renewable energy and geopolitics: a review. Renew Sustain Energy Rev 2020;122:109547. https://doi.org/10.1016/j.rser.2019.109547.